Zero Leverage Puzzle: Do Labor Laws Matter?

Hamid Boustanifar
EDHEC Business School

Arnt Verriest
KU Leuven

Forthcoming at *European Financial Management*

Abstract

We show that part of the zero-leverage puzzle in the US is explained by labor protection laws. We exploit the staggered passage of the good faith exception laws from 1967 to 1995 that increased the cost of firing workers. Using a difference-in-differences framework, we find that higher labor adjustment costs increased the likelihood of observing zero leverage firms by 22%. Our findings are robust to controlling for time varying firm and state characteristics, firm fixed effects, and any time-varying industry level shocks. We also show that these labor laws had significantly larger effects on zero leverage policies in states with stronger unionization, in industries with higher volatility and concentration, and in firms with higher labor intensity. Within-firm changes in debt policies and higher propensity of newer firms to be debt-free following stronger labor laws are almost equally important in explaining the evolution of zero leverage firms. The relationship between labor protection and *average* firm leverage shown in the prior literature is predominantly driven by zero leverage firms. Overall, our results complement the recent literature documenting and investigating the determinants of zero leverage puzzle by providing a novel explanation-- increased labor adjustment costs.

Keywords: Zero leverage, Net debt, Employee protection, Labor Protection

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1. INTRODUCTION

Recent research has documented that a substantial fraction of firms around the world are run with zero or almost zero leverage (more than 20% in the US, for example), a phenomenon called zero-leverage puzzle. This puzzling fact has been shown by Strebulaev and Yang (2013) for the US, Dang (2013) for the UK, Huang et al. (2017) for China, and Bessler et al. (2013) and El Ghoul et al. (2018) for a large sample of countries around the world. This phenomenon is typically referred to as a puzzle since, as argued in Strebulaev and Yang (2013), almost no model of capital structure predicts such a high proportion of firms with zero or almost zero leverage.[1] The extant literature trying to explain this phenomenon has highlighted the role of CEO ownership and family firms (Strebulaev and Yang 2013), macroeconomic conditions and financial constraints (Dang 2013), credit constraints (Devos et al. 2012), and the real option value of having debt (Lotfaliei 2018). Differences in the degree of zero leverage phenomenon across countries have been shown to be related to culture and trust (El Ghoul et al. 2018), creditor protection, legal origin, and dividend taxation (Bessler et al. 2013).

In this paper, we argue and provide evidence that employee protection laws can also explain part of the zero-leverage puzzle. While prior literature has documented a link between higher employee protection and lower average firm leverage (Simintzi et al. 2014; Serfling 2016), it is important to note that variables that affect (average) leverage may not have any power in explaining zero leverage puzzle. For example, while Berger et al. (1997) find that managerial entrenchment leads to lower leverage, Devos et al. (2012) show that zero leverage policies are not driven by entrenchment. Similarly, while industry and size have been shown to

[1] Baker et al. (2020) model rational leverage choices in the presence of mispricing risk. Within this (behavioral) framework, some firms rationally choose very low or very high leverage.

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be very important and robust determinants of firm leverage (Frank and Goyal 2009), Strebulaev and Yang (2013) find that "neither industry nor size can account for zero-debt policies". Indeed, this is the main reason why a growing literature cited above aims at understanding which factors explain zero leverage policies, despite about seven decades of research on capital structure that has shown important determinants of firm leverage. Overall, whether labor laws have significant explanatory power for zero leverage puzzle and what fraction of the documented link between labor laws and firm leverage is driven by zero leverage firms are open empirical questions that we address in this paper.

In an economy where firms face a high cost of firing employees, labor expenses are unavoidable in the short run as firms will not be able to discharge employees at a low cost in case of a temporary economic downturn. In such an economy, firms will have higher *operating* leverage since labor cost is not only a variable cost but rather has a fixed cost component.[2] Higher operating leverage makes firms riskier and as a result firms should respond with adopting conservative financial policies (Aiyagari 1994). Put differently, an increase in operating leverage induces the requirement for more financial flexibility, to offset the increased risk and to reduce the probability and costs of business termination or bankruptcy. Therefore, we predict that increases in employee protection should result in a higher likelihood of observing zero or almost zero leverage among firms.[3]

Empirically, investigating the impact of labor protection laws on adopting zero leverage policies is challenging due to endogeneity and omitted variable issues. For example, any cross-

[2] To be clear, the idea that labor cost is not variable but quasi-fixed has been documented in the prior literature (Oi 1962; Hamermesh and Pfann 1996; Boustanifar 2014). Here we are exploiting a law that *increases* the fixed-cost component of labor.

[3] The notion of operating leverage being traded off against financial leverage is supported in the literature (Gamba and Triantis 2008; Morellec et al. 2014).

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sectional correlation between labor protection and the zero-leverage phenomenon would be subject to criticism that regions with high labor protection are fundamentally different from those with low labor protection and the differences in tendency for adopting zero leverage may be the result of those (unobservable) characteristics and not labor protection laws themselves. Using changes in labor protection laws within a country would also be subject to the fact that the estimated effect cannot isolate the effect of any other macro shocks that affects all firms in the economy (access to finance, growth prospects etc.) around the change in labor laws.

To overcome the empirical issues mentioned above, we exploit the introduction of the good faith exception, one particular law in the Wrongful Discharge Laws (WDLs), which constitutes a drastic change in labor protection across US states. The good faith exception prevents employers from firing people out of bad faith, malevolence or retaliation. This exception has been interpreted quite broadly and its adoption is often considered as the strongest increase in employee protection, compared to other protection laws such as implied contract exception and the public policy exception (Autor et al. 2007).[4] An important feature of these laws was that they happened in a staggered way across the states. The fact that different states adopted good faith exemption laws in different years provides us with an excellent laboratory to test the effect of labor protection laws on the likelihood of adopting zero leverage policies by firms in a Difference-in-Differences framework. In other words, we can estimate the *change* in probability of firms adopting zero leverage policies after the law *within* the same state, controlling for similar change in states that did not pass the law. The treatment (control) group includes firms that are located in states that have adopted the good faith exception (are located in non-adopting states). Of course, an important assumption behind our analysis is that the regulatory changes across states were exogenous to zero

[4] We provide detailed descriptions of different employment protection laws in the US in Section 2.

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leverage phenomenon, in presence of our control variables. We provide multiple tests and analyses, and virtually all of them support this assumption. Most of our analyses are at the firm level and robust to inclusion of industry, year, state and firm level fixed effects.

Our main findings indicate that the good faith exception is positively associated with the incidence of having zero leverage and negative net debt (hereafter negative debt or ND). These effects are not only statistically significant but also economically meaningful. Economically, our results suggest that the introduction of the good faith regulation increases the probability of observing zero leverage firms by 3.2 percentage points or 22%. In addition, and importantly, we find that the positive documented link between higher employee protection and lower leverage in prior literature is predominantly driven by the zero leverage and negative debt firms. More precisely, after controlling for these firms, employment protection has no (statistical of economic) significant effect on firm leverage.

If as we predict, labor laws affect the decision to be zero levered through the labor adjustment channel, we expect predictable cross-sectional variation in the proportion of zero leverage firms across the population. First, our results should be stronger in states with low unionizations since firms in high unionized states are most likely faced higher firing costs even in absence of the Good faith laws. Second, we expect a larger effect in industries with higher volatility and those with high competition since firms in these industries are the ones that are more likely to face negative shocks and need more flexibility. Finally, since our argument is based on labor adjustment costs, we expect that our effects should be focused mostly on firms with relatively high labor intensity. We show that the effect of labor protection laws on zero leverage varies across states, industries, and firms in a theoretically predictable way that are consistent with our main arguments. These results increase our confidence that the association we are documenting is indeed

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the result of labor protection laws and is unlikely to be due to other confounding factors. In other words, any other alternative explanation should also be consistent with these heterogeneous effect of labor laws on different states, industries and firms.

Finally, we conduct a battery of additional analyses to investigate the robustness of our results as well as better understanding the mechanism behind the effect we document. Specifically, we show that the effect on labor laws on zero leverage policies is not driven by one influential state. Moreover, the statistical significance and economic magnitude of the effect barely change when we use alternative regression models such as probit or linear probability models or when we run regressions at the state-level. We also show that the results are robust when we use the full set of industry-year fixed effects to control for any unobserved time-varying industry-specific factor that could be correlated with both our measure of labor law and zero leverage policies. Finally, we find that within-firm changes in policies in favor of zero leverage and negative debt (i.e., when we add firm fixed effects) is responsible for between 30% to 50% of the overall effect we document.

Our findings explained above help to further understand the zero-leverage puzzle and its determinants. Previous papers have highlighted the role of family firms, CEO ownership, culture, trust, dividend taxation, creditor protection, macroeconomic conditions and financial constraints (Devos et al. 2012; Bessler et al. 2013; Dang 2013; Strebulaev and Yang 2013; Bigelli et al. 2014; Huang et al. 2017; El Ghoul et al. 2018; Lotfaliei 2018).[5] To the best of our knowledge, we are the first to show that stronger labor laws can also explain part of the evolution of zero-leverage puzzle.

Furthermore, we also contribute to the literature that investigates the impact of labor laws on different corporate policies such as takeover activities, leverage, investment and sales growth,

[5] The low-leverage puzzle – the fact that firms have on average lower leverage ratio than what is implied by most capital structure models- (Graham 2000) has also been shown to be driven mostly by zero-leverage firms (KORTEWEG 2010; Strebulaev and Yang 2013).

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and cash holdings (Simintzi et al. 2014; John et al. 2015; Serfling 2016; Dessaint et al. 2017; Bai et al. 2019; Beuselinck et al. 2021). The closest paper to ours is Serfling (2016) who shows that higher labor protection leads to lower firm leverage. As discussed earlier, an association between a variable and lower firm leverage does not suggest that the variable has explanatory power for zero leverage puzzle. We show that stronger labor protection is indeed an important determinant of zero leverage puzzle. In fact, we show that the relationship documented in Serfling (2016) disappears when we control for zero leverage firms.

The rest of the paper is organized as follows. In Section 2 we discuss the institutional background on labor protection. In Section 3 we describe our empirical design. In Section 4 we explain our data and sample. Section 5 presents our empirical findings and further analyses and robustness checks. Finally, Section 6 concludes our study.

2. Institutional Background on Labor Protection

In the common law free market identity of the U.S., it was often assumed that employers could discharge employees for good reasons, bad reasons, or no reason at all. This notion of "at-will" employment concerned both employers and employees. The labor contract was terminable at each party's request. In the 19th century the Supreme Court and the U.S. judiciary resisted federal and state intervention in regulating the labor market. Involvement by governmental agencies was considered to be an undesirable intervention on the economic freedom of businesses. Regulation was thought to be unnecessary and to be "unreasonable, unnecessary, and arbitrary interference with the right and liberty of the individual to contract" (Bernstein 2005).

Starting in the twentieth century, the legal environment gradually changed the nature of the employer-employee relationship. Employees began to become more and more protected

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against being discharged *unfairly*. For instance, the National Labor Relations Act of 1935 allows employees to be unionized and be involved in collective bargaining (Gorman and Finkin 1981). Unionization triggered the instalment of "just cause" amendments in employee contracts in order to protect against unfair dismissals (Abrams and Nolan 1985). Starting in the 1970s many states also started to install exceptions to the discharge at-will rule. These laws are referred to as the Wrongful Discharge Laws (WDLs). The WDLs consist of three common law exceptions: the good faith exception, the implied contract exception and the public policy exception. These exceptions afforded employees various means and levels of protection. Each state decided which exceptions it would recognize: the public policy exception was recognized by 42 U.S. states, the implied contract exception was recognized by 36 U.S. states, and the good faith exception was recognized by 14 U.S. states. The Appendix A1 provides a description of these three laws.

The WDLs have been the subject of investigation in various academic research papers in law, economics and finance, showing they affect various economic outcomes. Rather than providing an entire literature review, we focus on a few studies that are relevant to our current study. Autor et al. (2006) show that the passage of these laws reduced state level employment by 0.8-1.7%. Next, Dertouzos and Karoly (1992) report that the adoption of WDLs is equivalent to a 10% tax borne by employers. John et al. (2015) examine state variation in labor protections in an M&A setting, and find that shareholder returns are lower when the acquirer is from a state with strong labor rights. Focusing on a large sample of banks, Bird and Knopf (2009) report that the adoption of the implied contract exception increases labor expenses, and has a negative effect on profitability, once accurately controlled for the state's economic conditions. Although our focus is on these law changes in the US, other economies across the globe have introduced similar regulatory increases in labor protection, particularly in the OECD. For instance, Bryan et

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al. (1999) argue that labor mobility in the U.S. have enabled a quicker expansion of new industries than was the case in Europe or Japan. Bentolila and Bertola (1990) argue that the high level of employee protection in European labor markets adds to the weak performance of European economies at the end of the 20th century, referring to these countries as "Eurosclerotic" economies. Many of these studies' results imply that higher employment protection and more rigid labor markets negatively affect factor productivity and firm profitability, as well as impede their operational flexibility.

However, other studies point at the benefits for society of an enhanced employee protection (Blank 1994; Saint-Paul 2002). Some studies have also detected beneficial effects at the firm-level. Acharya et al. (2014) argue that employment protection enables employees to take risks and undertake innovative projects, and subsequently find a positive relation between employee protection and the success of innovation activities.

In this study, we focus on the state adoption of the good faith exception. The introduction of this law is the most far-reaching of the three exceptions in terms of enhanced employee protection. The good faith exception refers to an implicit assumption of good faith and fair treatment between employees and employers. This law prevents employers from firing people out of bad faith, malevolence or retaliation. This exception has been interpreted quite broadly and its adoption is often considered as the strongest increase in employee protection of the three exceptions (Autor et al. 2007; Serfling 2016).

Our identification assumption in this paper is that the passage of these exceptions, and in particular the good faith exception, to the at-will employment doctrine is exogenous to firms' financial policies and in particular whether they are zero leverage or not. Prior research documents that the passage of these laws is unaffected by firm and state-level variables (Acharya

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et al. 2014), and is in fact more exogenous than firm-level unionization decisions (John et al. 2015). Furthermore, Section 5.1 below directly addresses the endogeneity of the passage of these laws in several ways and shows that there is no evidence that zero leverage laws or a variable correlated with that can predict the passage of employment protection laws. Therefore, we exploit employee protection legislation as an exogenous event to firms' decision to adopt zero leverage policies. More specifically, we use the good faith exception adoption occurring at the state level as an exogenous shock to employee protection and test whether labor protection laws can help explaining part of the zero-leverage puzzle. We next proceed to empirically examine our conjectures.

3. EMPIRICAL STRATEGY

Our main empirical methodology is a difference-in-differences model where the unit of observation is firm-year. The main regression equation is as follows:

$$ZL_{ist} = \alpha + \beta \text{Good Faith}_{st} + \gamma \text{Firm Controls}_{ist} + \delta \text{State Controls}_{st} + \text{State Fixed effects} + \text{Year Fixed Effects} + \text{Industry Fixed Effects} + \varepsilon_{ist}$$

where $i, s,$ and t stand for firm, state, and year. Our dependent variable, ZL_{ist}, is a dummy variable indicating whether the firm $i,$ in state $s,$ and at year t is a zero leverage firm or not. We employ two principal indicators for zero leverage. The first, *Zero Leverage,* following Strebulaev and Yang (2013), equals 1 when the sum of book value of long-term debt (*dltt*) and short-term debt in current liabilities (*dlc*) is less than 5 percent of total assets, and zero otherwise. The second, *Negative Debt*, equals 1 if the sum of book value of long-term debt (*dltt*) and short-term debt in current liabilities (*dlc*) less cash and cash equivalents (*che*) is negative, and zero otherwise.

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Good Faith, our main variable of interest, is an indicator equal to 1 for firms headquartered in states that have adopted this law as of year t, zero otherwise. Indicator variables for adoption of the implied contract and public policy exception are also included. We estimate the model using a logit model. The main coefficient of interest is β, which is in fact the diff-in-diff estimator. It estimates the changes in the probability of being a zero leverage firm in treated states (states that changed their laws), controlling for the aggregate trends in zero leverage which are not due to these law changes.

Firm Controls is a set of firm specific control variables that could be related to zero leverage policies (or firm leverage in general). We include the logarithm of total assets (*Log Assets*) as a measure of firm size. Growth opportunities is captured by market-to-book measured as firm equity value normalized by its book value (*Market-to-Book*). Profitability is measured as net income before extraordinary items scaled by total assets (*Profitability*). Capital intensity is property plant and equipment scaled by total assets (*Capital Intensity*). Dividend payments are measured as a dummy variable if a firm pays dividends or not (*Dividend Payer*). Operating volatility is captured by the standard deviation of operating cash flow (*Cash Flow Volatility*). Capital expenditures are measured as the level of yearly investment scaled by assets (*Capex*). Firm age is the log of the number of years the firm is covered by Compustat (*Log Age*). Working capital is current assets minus cash scaled by current liabilities (*Working Capital*). All firm-specific control variables are winsorized at the top and bottom percentile, with the exception of *Log Assets*, *Log Age* and *Dividend Payer*.

We also add time-varying state controls to capture time varying state level factors between states (*State Controls*) that could potentially be correlated with changes in labor laws and zero leverage decisions. The model includes state GDP growth and state GDP per capita (S*tate GDP*

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Growth and *State per capita GDP*). Autor et al. (2006) argue that if a state in the same circuit adopts the good faith exception, there is a higher likelihood the home state will also adopt it. Therefore, we include an indicator that captures this notion (*Circuit States' Good Faith*).

Importantly, we include the full set of state, year and industry fixed effects to our regressions. State fixed effects control for any time-invariant differences across states such as persistent differences in access to credit or stable differences in types of firms or businesses across states. Year fixed effects captures any aggregate macro shocks and trends such as overall changes in business characteristics, globalizations, technological changes, and so on. Since *Good faith* indicator has state and year dimension, the inclusion of states and year fixed effects ensures that our estimated β coefficient is driven by *within state changes* of zero leverage firm in good faith laws, controlling for changes in zero leverage firms in other states. In addition, the inclusion of industry fixed effects ensures that our results cannot be attributed to any persistent differences across industries such as the tendency of firms in particular industries to have zero leverage. We estimate the model using logit regressions in the main analyses and report standard errors clustered at state level, which is the most conservative way of clustering in these types of setups (Bertrand et al. 2004).

It is important to note that our empirical setup is a generalization of standard diff-in-diff methodology since the changes in laws are staggered during our sample period. This means that our control states are not those states that never changed their laws. Instead, our control groups at year t are those states that did not change their law in year t, even though they will change later on. The fact that our treated states have been part of control group before they change their laws reduces the concern that treated states (those that they change their laws) may be completely different from control states (those that never changed their laws).

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We acknowledge that there could still be *unobservable time-varying* state level variables that correlate with the timing of good faith exception laws across states and at the same time affect the tendency of firms to be run with zero leverage. While we cannot include state-year fixed effects in our regressions since our main variable of interest varies at state-year level, we perform further analyses and show that the strength of the effect we document varies across firms, industries, and states in a predictable way if the effect is indeed due to labor laws. These results diminish potential concerns related to interpretation of our results.

4. DATA

4.1 Sample Selection

In this study we consider exceptions to the at-will employment doctrine and examine the adoption of Wrongful Discharge Laws (WDLs). These laws encompass the good faith, implied contract, and public policy exceptions and are also considered in other studies including Acharya et al. (2014) and John et al. (2015). We conduct our analysis at the firm level. Our sample includes all firm-year observations available in Compustat between 1967 and 1995. The start is 1967, five years before California adopted the implied contract exception in 1972. The sample ends in 1995, as by then most passages of the WDLs occurred. Extending the sample with 5 years does not change our main inferences. Appendix A.2 exhibits the adoption timing of these three laws. We exclude regulated utilities (SIC 49) and financial firms (SIC 60-69) as well as observations with negative sales (*sale* < 0) and those for which there is no data for debt (*dltt* and *dlc*), cash and cash equivalents (*che*) and total assets (*at*). We have a total sample size of 134,177 firm year observations.

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4.2 Summary Statistics

Descriptive statistics are shown in Table 1. The average proportion of firms with zero leverage and those with negative debt in our sample are 17.6% and 27.2%, respectively. We find that the good faith adoption firms represent 16.6% of the sample, from 14 out of 50 states (see Appendix A.1 for details). These proportions are much higher for public policy and implied contract observations, both at about 50% of observations. The average firm has much a much higher amount of assets than the median firm indicating a highly positively skewed distribution of firm size. The median firm has positive earnings (*Profitability*), although more than a quarter of the sample reports a loss. The average firm has about a third of their assets in property, plant and equipment (*Capital Intensity*).

Figure 1 shows the aggregate evolution of zero leverage and negative debt firms during our sample together with the accumulated number of states that have adopted Good Faith laws. As shown, the proportion of zero leverage firms doubled during our sample period (from 13% in 1970 to 26% in 1995). We observe similar magnitude of increase for proportion of firms with negative debt that went from 20% in 1970 to about 37% in 1995. The correlation between proportion of zero leverage and negative debt series (aggregate) is 97%. Interestingly, as shown, the accumulated number of the states that have adopted the law has a high correlation (94%) with the proportion of zero leverage (or negative debt) firms. Overall, we confirm that the aggregate proportion of firms with extremely low leverage increased substantially during our sample and their aggregate evolution over time is quite similar to aggregate number of states that have adopted the law.

However, we note that our paper is not meant to explain this aggregate trend that could be explained by different macro shocks (technological change, changes in nature of businesses etc)

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to the economy. As explained before, our empirical methodology controls for all macroeconomic or aggregate shocks that could even be specific to each industry in each year, and focuses on changes in the proportions of zero leverage firms within treated states after controlling for similar changes in control states. In other words, our results give an estimate of how much labor laws help explain zero leverage phenomenon above and beyond the overall trends in zero leverage firms.

In Panel A of Table 2 we show the proportions of zero leverage and negative debt firms per state. For good faith adopting states, we show the numbers before and after the adoption. Panel B of Table 2 summarizes these proportions (analyses are done at the state-year level) and test results for pair-wise differences between good faith adopting and non-adopting states. Interestingly, we find that, before the adoption of the good faith exception, there is no significant difference in the proportion of zero leverage and negative debt firms between non-adopting states and adopting states (for zero leverage, 14.6% vs 14.7% and for negative debt, 27.1% vs 28.1%). However, we do find significant differences between non-adopting and adopting states *after* the adoption of the good faith exception: 20.8% for zero leverage and 36.6% for negative debt are significantly higher proportions than the ones for non-adopting states (which have 14.6% and 27.1%, respectively). We also find that the proportions of zero leverage and negative debt firms in good faith adopting states has increased significantly after adoption. These univariate state-level results are consistent with our prediction that labor laws explain part of the zero-leverage phenomenon.

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5. EMPIRICAL RESULTS

5.1 Determinants and Endogeneity of Labor Laws

One important identifying assumption behind our empirical methodology is that the treatment (adoption of good faith laws) was not driven by our dependent variable (zero leverage policy). For example, if states adopted the laws when the proportion of zero leverage firms was going up, we may incorrectly associate the increase in zero leverage firms to the law change. Therefore, it is important to investigate the determinants of the adoption of good faith laws. In particular, it is critical to know whether firm's financial policies (and specifically zero leverage policies) affected the timing of law changes. For instance, one can imagine that when the local economic or credit conditions within a state deteriorate, more and more firms will have limited access to credit and will be run with zero leverage and that could also determine the choice of states to strengthen their labor laws. In this case, the increased in zero-leverage policy firms is not due to changes in labor laws but instead it is the result of local changes in economic conditions.

We address this issue in three different ways. First, we start by examining whether the average proportion of zero leverage firms in a state helps predict the timing of the adoption of good faith laws. Figure 2 shows that the proportion of zero leverage (Panel a) and negative debt firms (Panel b) before the law cannot predict the timing of the adoption of good faith laws across states. The t-statistics for the correlation of the two variables is -0.09 and -0.07 in Panel (a) and (b), respectively.

Second, we run a hazard model to test whether zero leverage policies (or other firm relevant financial conditions and policies) influence the likelihood that a state adopt the good faith exception laws, given that it has not adopted yet. *Table 3* reports the results. We do not find any

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evidence that average proportion of zero leverage firms, states' income, states' income growth, unemployment rates or unionization can help predicting the timing of the passage of good faith laws. We do, however, find some (weak and not robust) evidence that *public policy* and whether other states in the same legal circuit have adopted the good faith exception are correlated with the timing of the law adoptions. Therefore, we control for these in all our regressions.

Third, we run a set of placebo regressions where we randomly draw adoption years with replacements from the empirical distribution of actual adoption years. Using this placebo dates, we run our main regressions for 100 times and retrieve the point estimate on our good faith law variable. The average of these estimates are 0.003 and only in 2 percent of times the null hypothesis (that the coefficient is zero) is rejected.

In sum, our graphical investigation as well as our hazard model and the placebo test all suggest that the timing of adoption of good faith laws was exogenous to zero leverage policies.

5.2 Labor Laws and Zero Leverage

In this section, we investigate the effect of adoption of good faith laws on the probability of firms following a zero leverage policy. We conduct firm-level logit specifications, in which the dependent variable is an indicator equal to one if the firm is zero leverage in a particular year. Each specification includes industry, year and state fixed effects, and the reported standard errors are clustered at the state level. *Table 4* presents our results. In Panel A, the dependent variable is the indicator for zero leverage. Column (1) shows baseline regressions with the three labor law indicators. The coefficient on *Good Faith* is positive and statistically significant, while *Implied Contract* and *Public Policy* are not. Column (2) adds firm-specific controls, with similar results. *Good Faith* is positive and highly significant, in line with the notion that post good faith adoption,

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a larger proportion of firms maintain a zero leverage policy. Similar inferences can be drawn from columns (3) and (4), where we add additional firm characteristics decreasing the number of observations available.

Many firm characteristics are significant. We find that larger, older and more capital intensive firms as well as firms with higher capital requirement needs are less likely to be zero leverage, in line with findings in prior research and our expectations. Conversely, more profitable firms and those with higher growth opportunities are more likely to follow a zero-leverage policy. In column (5), which constitutes as our main and full specification, we add state level control variables, of which only GDP growth is significant.[6] Again the estimated coefficient on *Good Faith* indicator is positive and statistically significant. The economic magnitude of the effect is also sizable. For example, when converted to the probabilities, our results of column 5 (with the full set of control and fixed effects) suggest that following the adoption of good faith laws, the probability of observing a zero leverage firm in the state increases by 3.2 percentage points.[7] This is a sizable effect given that the unconditional probability of having a zero leverage firm is 14.7% before the law adoption (see Panel B Table 2). Put it differently, on average, the probability of observing zero leverage firms goes up by 22% (=3.2/14.7) in states following the adoption of good faith law.

In Table 4 Panel B, we test for our alternative dependent variable, *Negative Debt*, an indicator equal to one for firm observations with negative net debt, or cash larger than debt. Similar to Panel A, we find that firms are more likely to be run with negative net debt following the adoption of the good faith law (compared to the control group). As shown, this result is robust

[6] As we include many indicator variables in our binary dependent variable specifications, including dummies for state, year and industry, some observations are dropped due to a lack of within-group diversity on the dependent variable.
[7] It should be noted that the coefficients reported are raw logit regression estimates, not odds ratios or marginal probabilities.

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across different specifications. Importantly, the size of coefficients is stable across different specifications which suggests that *Good Faith* indicator does not have a high correlation with other controls and fixed effects we introduce, hence, supporting our identification assumptions. It is also important to note that the economic magnitude of the law is sizable. For example, the estimated coefficient of 0.325 on *Good Faith* indicator in Column 5 Panel B of Table 4 implies that the probability of observing firms with negative net debt increases by 5.7 percentage points following the adoption of good faith law. This is an increase of 20% (=5.7/28.1) compared to the average probability of negative debt firms before adoption of good faith exception.

In sum, our results show that the good faith exception to at-will employment has a positive and significant impact on the likelihood of observing zero leverage firms or those that maintain a negative debt policy. These findings are consistent with the notion that firms decide to be zero levered as a compensation for an increased, or expected increase in, operating leverage due to higher employee protection.

5.3 Employment Protection and Average Leverage: The Role of Zero Leverage Firms

In the previous section, we showed that stronger labor protection laws have significant explanatory power for zero leverage puzzle. As discussed earlier in the paper, this result is novel and could not be drawn from the findings of the prior literature that shows a negative link between stronger labor protection laws and average firm leverage (Serfling 2016).

Having said that, an interesting question would be whether zero leverage and negative net debt firms play an important role in explaining the effect documented in Serfling (2016). To investigate this, we run similar analysis as in Serfling (2016) where we use book and market leverage as our dependent variables. Columns 1 and 3 of Table *5* confirm the association: the adoption of Good faith laws reduced both book leverage and market leverage. However, when we

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control for zero leverage and negative debt firms in columns 2 and 4, the coefficient on Good faith is reduced significantly in size and becomes statistically insignificant. This suggest that the labor protection laws have a non-linear effect on firm leverage and the negative and significant effect between the laws and average firm leverage documented in the prior literature predominantly comes from zero leverage and negative debt firms. While our goal in this paper is to understand the role of labor protection laws in explaining zero leverage puzzle, investigating why labor protection laws have much lower impact on leverage of firms that are not zero-leverage (or do not have negative debt) is an interesting area for future research.

5.4 Dynamic Impact of Labor Laws on Zero-leverage

In the previous sections, we have shown that the adoption of good faith laws leads to higher average probability of firms adopting zero leverage policies, compared to before the adoption and controlling for trends in firms located in states that are not adopting these laws. In this section, we investigate the dynamic effect of changes in good faith laws on zero leverage policies. To this end, we estimate a year-by-year effect of law changes on zero leverage policy in a four-year window around the time of law change. Specifically, we run the following logit regression:

$$\begin{aligned} ZL_{ist} = \alpha + \beta_1 GF_{st}^{\leq -4} + \beta_2 GF_{st}^{-3} + \beta_3 GF_{st}^{-2} + \beta_4 GF_{st}^{-1} + \beta_5 GF_{st}^{+1} + \beta_6 GF_{st}^{+2} + \beta_7 GF_{st}^{+3} \\ + \beta_8 GF_{st}^{\geq +4} + \text{Firm Controls}_{ist} + \text{State Controls}_{st} + \text{ State Fixed effects} \\ + \text{Year Fixed Effects} + \text{Industry Fixed Effects} + \varepsilon_{ist} \end{aligned}$$

where instead of one dummy variable for good faith, we have a series of dummy variables indicating each year around the law changes. Specifically, GF_{st}^{L} for positive L (negative L) gets 1 if the firm is headquartered in a state that has adopted the law L years ago (will adopt the law in L

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years), and zero otherwise. We exclude the dummy variable corresponding to the year of law adoption, therefore all the β coefficients are estimated relative to the reference year, the year of adoption Good Faith law. We also cluster standard errors at the state-level.

Figure 3 shows the β coefficients together with the 95% confidence interval bound. The figure highlights two important messages: First, all the estimated coefficients before the law adoption are close to zero in magnitude, statistically insignificant, and there is no visible trend in the years prior to the law change. This strongly suggest that the trend in our dependent variable was similar across treatment and control states prior to the treatment (the parallel trend assumption). In addition, we observe that following adoption of the law, the likelihood of being zero leverage starts increasing and will become statistically significant two years after the law. The effect also persists in the following years. To sum up, these dynamic results show that while there was no significant difference in trends of zero leverage firms across treated and control groups prior to the adoption, the likelihood of being a zero leverage firm significantly increased compared to the control group following adoption of the law.

5.5 Heterogeneous Effect of Labor Laws on Zero Leverage

In this section, we conduct several additional tests to investigate conditions in which the effect of the good faith exception on the likelihood of zero leverage is stronger or weaker. Ideally, this battery of sensitivity tests may provide us with more insights into how labor regulation affects corporate finance decisions, and in this case leverage conservatism. Additionally, these tests may further provide indications that our main result is not driven by an unobserved time-varying state factor. We conduct four such tests exploiting cross-sectional differences in unionizations, industry volatility, industry concentration, and employee intensity.

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First, we investigate whether the effect of the good faith adoption on the likelihood of zero leverage is stronger in states where unionization is lower. Workers that are part of a union are generally less affected by the good faith exception (Autor 2003). Therefore, we expect the good faith exception to have a bigger effect on the likelihood of zero leverage in cases with less unionized employees. Data is gathered from the IPUMS-CPS database where we assess the percentage of unionized workers per state per year (*unionization*). We split the sample based on high versus low *unionization*.[8] The low unionization group contains observations from state-years below the median *unionization*, and the remaining observations go to the high unionization group. Table *6*Panel A summarizes the results. For brevity, we only report coefficients on *Good Faith*, our main variable of interest. Regression analyses and control variables are the same as those in specification 5 of Table 4. In line with our predictions, we find a larger coefficient on the good faith exception indicator in low unionization states.

Second, we test whether the effect of the good faith adoption on the likelihood of zero leverage is stronger in more volatile industries. We predict firms operating in more volatile environments to be more inclined to adopt a zero leverage policy in response to an increased labor rigidity, as there is a higher need for these firms to have a low financial risk. Therefore, we expect to find stronger effects in industries with more operating volatility. To capture *industry volatility*, we use the previously defined *cash flow volatility* and calculate the average over all firms per industry, per state, and per year. We split the sample in two groups, high versus low *industry volatility*, based on the median. Table *6*Panel B summarizes the results. In line with our predictions, we find strong results consistent with the notion that in response to the Good Faith exception, firms

[8] We also test these cross-sectional heterogeneous effects via interacting the variables (instead of splitting the sample) and report the results in Table Appendix A4. Overall, our results across the two methods are consistent, except for industry concentration.

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in more volatile industries are more likely to switch to a zero-leverage policy and maintain more cash than debt than firms in less volatile industries.

Third, we examine whether the effect of the good faith adoption on the likelihood of zero leverage is stronger in less concentrated industries. Firms operating in more competitive industries are more likely to see their operational successes turn or to lose their competitive edge more quickly. Industry concentration is the sum of squared relative sales, measured as each firm's sales divided by the industry's sales. Insofar as industry concentration captures the degree of competition in a certain industry, we predict to find stronger effects in less concentrated industries. Our results, shown in Table *6* Panel C, indeed show evidence consistent with this prediction. For zero leverage, we find a significant impact of good faith but only in the less concentrated, and therefore likely more competitive, industries. For negative debt, we find a significant result in both subsamples, although the effect in the less concentrated subsample is larger than in the other subsample.

Fourth, we investigate the moderating effect of the extent to which firms rely on employees (i.e. firms' employee intensity). An increase in employee protection is expected to affect labor-intensive firms more severely. Therefore, we expect the good faith adoption to urge managers to be zero levered especially for firms that rely a lot on employees and for those that have high salary expenses. We split the sample based on labor intensity which we measure by calculating the ratio of number of employees (in thousands) over sales (in millions of dollars).[9] Table *6*Panel D shows results for these tests. In line with our expectations, we find a significantly positive coefficient on *Good Faith* for the sample of firms with high labor intensity. Coefficients on *Good Faith* for the

[9] Alternatively, one could measure labor intensity as total labor cost as proportion of sales. However, data on staff expenses are missing for almost 90% of firm observations, while number of employees is usually available.

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other half of the sample are still positive, although not always statistically different from zero, and but its magnitude is much smaller. The Chow-test shows the difference between the coefficients on *Good Faith* in the low versus high labor intensity group. We find that the coefficient in the high employee intensity group is significantly larger than in the low employee intensity group.

Collectively, the results from these four cross-sectional tests reported in Table *6*show that the effect of employee protection shocks on the likelihood of firms following zero leverage policies is stronger when unionization is lower, in industries with higher volatility, and in firms with more labor-intensive operations. The fact that exploiting cross sectional differences across firms, industry, and states gives us heterogeneous effects in a theoretically predictable way significantly diminishes the potential concern that some unobserved time varying factor is driving our results. In other words, any alternative explanation for our results needs to generate consistent predictions about the heterogeneous effect of the laws on unionization, industry volatility, industry concentration, and labor-intensive firms.

5.6 Additional Analyses and Specifications

Influence of Each Single State

In the first set of robustness checks, we would like to examine the sensitivity of our results with respect to exclusion of observations from different states. In other words, we would like to understand whether the effect of good faith exception laws on zero leverage we have documented is due to one influential state or not. To do so, we re-run the specification 5 from Table 4 fifty-one times, dropping observations from a different state in each run. Figure 4 reports the results for both of our dependent variables and shows for each run the magnitude of the coefficient on the *Good Faith* indicator, represented by the dots, and its z-stat, represented by the triangles. In subfigure

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(a), testing the likelihood of zero leverage, we find that the size of the Good Faith coefficient (the dots) is stable and does not meaningfully differ across each specification. Specifically, the coefficient of interest ranges between 0.233 (when leaving out Illinois) and 0.343 (when leaving out Massachusetts), which are not far from the average coefficient of 0.280 we estimated for the entire sample. Importantly, when leaving out Californian observations, which effectively means dropping almost half of treated firm observations, the resulting estimated coefficient (0.260) is very close to the estimated average coefficient of 0.28. However, as expected, its significance level does decrease (z-stat of 1.92, significant at the 10% level).

We find similar results in subfigure (b) of Figure 4, which shows the estimated coefficients and their respective z-stat of Good Faith indicator on negative net debt firms. As shown, the magnitude of estimated coefficients of Good Faith when dropping observations from different states is quite stable and close to the full sample coefficient of 0.325. The largest deviation happens when we exclude California, in which case the estimated coefficient of interest is 0.233 but remains economically sizable and statistically significant at the 5% level (z-stat of 2.47). Overall, we conclude from the results of these 51 regressions illustrated in Figure 4 that our documented results are not attributable to one particular state.

Alternative Regression Models

All the main results we have reported so far showing the impact of good faith exception laws on the likelihood of zero leverage or negative net debt firms are the outcome of logit regressions. In this section, we investigate the robustness of those results when we use alternative models that are also frequently used. Specifically, we use our full regression specification with all

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set of controls and fixed effects (Column 5 of Table 4) and run it using linear probability model and probit regressions (instead of logit).

Table 7 reports the impact of good faith exception law on the likelihood of zero leverage and negative net debt firms using linear probability (Columns 1 and 2, respectively) and probit (Columns 3 and 4) models. We note that all regressions include state, year and industry fixed effects as well as our full set of control variables. As shown, the estimated coefficients on *Good Faith* indicator are all positive and statistically significant across different models and dependent variables. More importantly, the economic magnitudes of the effects are also sizable and consistent with those presented before using logit regressions. For example, the coefficient of 0.043 and 0.061 in Columns 1 and 2 imply that the probability of observing zero leverage and negative net debt firms goes up by 4.3 and 6.1 percentage points (compared to 3.2 and 5.2 in logit regressions), respectively, following adoption of good faith exception law. The corresponding magnitudes in probit regressions are estimated to be 3.6 and 6 percentage points. Overall, the results presented in Table 7 show the robustness of the effect of good faith exception laws (both statistically and economically) in explaining zero leverage phenomenon across different regression methods.

The Role of Industry (Persistent or Time-varying) Factors

In all main regressions presented so far, we have included industry fixed effects to control for *any* observable and/or unobservable persistent differences across industries that could be correlated with our dependent variable. However, one might be concerned that the relation between good faith exception laws and zero leverage policies we document is purely due to some *time-varying unobserved* industry characteristics. This concern is of course a legitimate one if the time-varying unobserved factor is related to zero leverage policies *and* is correlated with *Good Faith*

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indicator. While we cannot think of a plausible factor that satisfies the abovementioned conditions, one may definitely exist.

To address the role of industry factors (both constant and time-varying) in our context, we first go one step back and run our main regression specification with all controls, state and year fixed effects but excluding industry fixed effects. We then introduce industry fixed effects in the regressions to understand the role of persistent industry factors in explaining the relationship between *Good Faith* and zero leverage policies. In the next step, we introduce the full set of industry-year effects to control for any observed or unobserved time-varying industry-level factor that could explain the relation between good faith exception laws and zero leverage policies. These factors could be any time-varying macroeconomic or aggregate shocks (e.g. financial development, changes in technology) that are industry-specific or any potentially time-varying industry-specific factor such as higher propensity of firms in some specific industries to have zero-leverage or the change in that propensity across industries.

Table 8 shows the result of these regressions.[10] The dependent variable in the first three columns is the zero-leverage indicator. Column 1 includes state and year fixed effects, Column 2 adds industry fixed effects and Column 3 has the full set of industry-year fixed effects allowing for time-varying industry factors. As shown, the coefficient of *Good Faith* changes very little across the three specifications and goes from 0.044 in the first specification to 0.043 and 0.035 in the second and third specifications, respectively, when we introduce industry and industry-year fixed effects. Moreover, the coefficient of interest remains statistically significant across all three specifications.

[10] From now on, we report results from linear probability models in that the interpretation of estimated coefficients are straightforward and we have already shown that this model yields similar results as those of logit and probit. All our results are robust when we use the logit model.

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We find similar results in regressions of negative debt (Columns 4-6). Specifically, in these regressions the estimated coefficient on *Good Faith* goes from 0.066 in the specification without industry fixed effects to 0.061 and 0.040 in models with industry and industry-year effects, respectively. All these coefficients retain their statistical significance across specifications.

Overall, the results presented in Table 8 show that the effect of Good Faith on zero leverage and negative debt documented in this paper cannot be attributed to any industry-level (observed or unobserved, fixed or time-varying) factor. To be clear, these results do *not* suggest that industry factors are irrelevant in explaining the zero leverage phenomenon. In fact, we do see about 20 percent increase in r-squared when we add industry fixed effects, which suggests that industry factors do matter. However, the stability of *Good Faith* coefficient before and after inclusion of industry effects shows that industry factors (even though important in explaining part of zero leverage puzzle) cannot explain the impact of adoption of good faith exception laws on the likelihood of observing zero leverage firms.

Within-Firm Effects

We have documented that following the adoption of good faith exception laws by states, the probability of observing firms with zero leverage or negative debt increases. Such an effect could come from either (a) higher tendency of existing firms changing to adopt zero leverage or negative debt, (b) business dynamics that favors zero leverage, i.e. higher creation and lower death rate of zero leverage firms, or (c) both of the above. To shed light on this point, we run our full model specification for both of our dependent variables, but with firm (rather than industry) fixed effects.

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The results are shown in Table 9 Column 1 (for zero leverage) and Column 3 (for negative debt). The coefficient of *Good Faith* implies that within the same firms, the probability of zero leverage and negative debt increased by 1.4 and 3 percentage points, respectively, following the adoption of good faith exception laws. As reported in Table 7 Column 1 and 2, without firm fixed effects, the corresponding probabilities were 4.3 and 6.1 percentage points. This means that within-firm changes towards adopting zero leverage policy explains 32% (=1.4/4.3) of the overall effect we have documented in this paper. For the negative debt policy, within firm changes following the law is responsible for 49% (=3/6.1) of the overall effect.

In Columns 2 and 4 of Table 9, we add the full set of industry-year fixed effects to control for time-varying industry factors. Doing so, increases the magnitude of *Good Faith* coefficient from 0.014 to 0.019, while decreasing the magnitude of the coefficient in negative debt regression from 0.030 to 0.026. Comparing these new coefficients with corresponding coefficients in regressions without firm fixed effects (Columns 3 and 6 of Table 8), the proportion of the effect attributed to within-firm changes is 54% (=0.019/0.035) and 65% (=0.026/0.040) for zero leverage and negative debt, respectively.

To sum up, our results in Table 9 suggest that within-firm changes to become a zero leverage (negative debt) firm is responsible for between 30% to 50% (50% to 60%) of the overall effect of *Good Faith* we have estimated. The remaining half could be due to either adoption of the law making it more likely for entrants to adopt zero leverage and negative debt policies or because of higher survival probability of zero leverage firms in those states.

State-Level Specifications

All our regressions so far have used firm level data to estimate the effect of states' adoption of good faith laws on the likelihood of firms run with zero leverage or negative net debt. Indeed,

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we prefer using firm-level data (rather than state-level) in this context since our dependent variable is a firm-level measure. In addition, using firm level data allows us to control for many variables and factors at the firm or industry level that could drive the decision of firms to be run with zero leverage or negative debt policies.

Having said that, since our *Good Faith* indicator changes at the state and year level, it is possible to run regressions at the state-level by aggregating our firm level data. The advantage of this method is that we get a continuous dependent variable which is the proportion of zero leverage or negative debt firms in a state in a given year. Therefore, OLS would be the most appropriate regression method, the coefficients have a straightforward interpretation, and it serves as another robustness check to ensure the consistency of our results across different methods and specifications.

Specifically, for our state-level regressions, we run the following model:

$$ZL_{st} = \alpha + \beta \text{Good Faith}_{st} + \delta \text{State Controls}_{st} + \text{ State Fixed effects} + \text{Year Fixed Effects} + \varepsilon_{ist}$$

where the dependent variable is the proportion of zero leverage (or negative debt) firms in state *s* at year *t*. We include state and year fixed effects to control for any state-specific factors (such as exposure to different industries or having different business policies) or year-specific effects (such as any macro shocks including technological changes). The coefficient of β above has a difference-in-differences interpretation, which measures the changes in the proportion of zero leverage (or negative debt) firms in states that adopt the Good Faith laws (treated group), controlling for changes in these dependent variables in states that did not adopt the law (control group). In addition to the above set of fixed effects, we also add state linear trends in an additional specification to control for trends such as increasing the importance of one specific industry within

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a state. We run the regression above using OLS while weighting each state observation by the average number of firms in that state during the sample period.[11]

Table 10 provides the results of our state-level regressions for the two dependent variables: proportion of zero leverage (Columns 1-3) and negative net debt (Columns 4-6) firms. The results are all in line with our previous findings that the adoption of good faith laws has a positive and significant effect on both zero leverage and negative debt policies. The magnitudes of the effects are also consistent with what we found in the firm-level regressions. Specifically, the results show that following the adoption of good faith laws, the proportion of zero leverage and negative debt firms went up by 4.5 and 6.6 percentage points in those states (compared to the control states). Given that the average proportion of zero leverage and negative debt firms before adoption of good faith laws were 14.7% and 28.1% (see Table 2 Panel B), our state-level results also show sizable effects which are also in line with firm-level estimates reported in previous tables.

6. CONCLUDING REMARKS

This study contributes to the recent literature investigating explanations for zero-leverage puzzle, a puzzling phenomenon that a significant portion of firms are run with zero or almost zero leverage. We show that stronger labor protection laws significantly increase the proportion of firms financed with zero leverage or negative debt. Specifically, we find that the staggered adoption of the good faith exception to at-will employment relationships increases the proportion of U.S.

[11] Weighting is needed since some states have very few firms. Therefore, the proportion of zero leverage firms in those states could jump dramatically following adoption of good faith laws due to only a few firms changing their policies. This could lead us, for example, to find a positive effect even though the law had no or very weak effect on states with very large number of firms. Therefore, weighting is required to get an estimate of the effect that is not driven by few very small states.

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corporations carrying no debt (or negative net debt) by 22% (in most conservative estimates). In addition, we show that the magnitude of this effect varies in a theoretically predictable way across firms, industries and states. We believe our main contribution is to highlight a novel channel (i.e. stronger labor protection laws) that is empirically significant in explaining the zero leverage puzzle.

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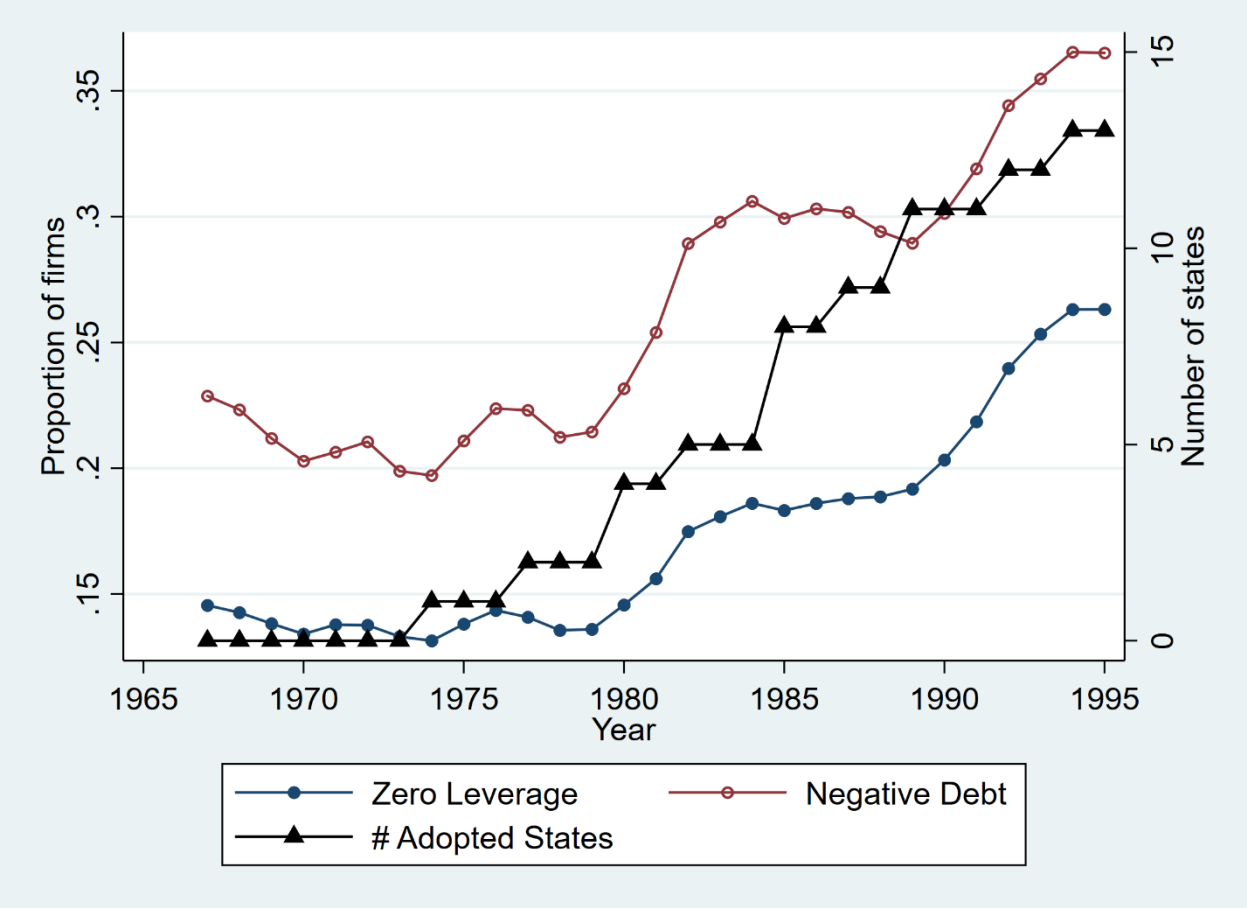


Figure 1. Evolution of Zero Leverage and Negative Debt Firms

This figure shows the proportion of firms with zero leverage and those with negative debt during our sample period (left axis) as well as the cumulative number of states with Good Faith adoption (right axis). The plotted statistics on proportion of zero leverage and negative debt firms are three-year moving averages.

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Figure 2. Good Faith Adoption Year and Pre-existing Zero Leverage Firms, by State

This figure shows the year of good faith adoption by each state against the (a) average proportion of zero leverage firms and (b) negative debt firms prior the adoption. Only states that have adopted the exception are plotted. The line shows best linear fit which is an outcome of a simple regression between the two variables. The corresponding t-stat for the correlations is -0.07.

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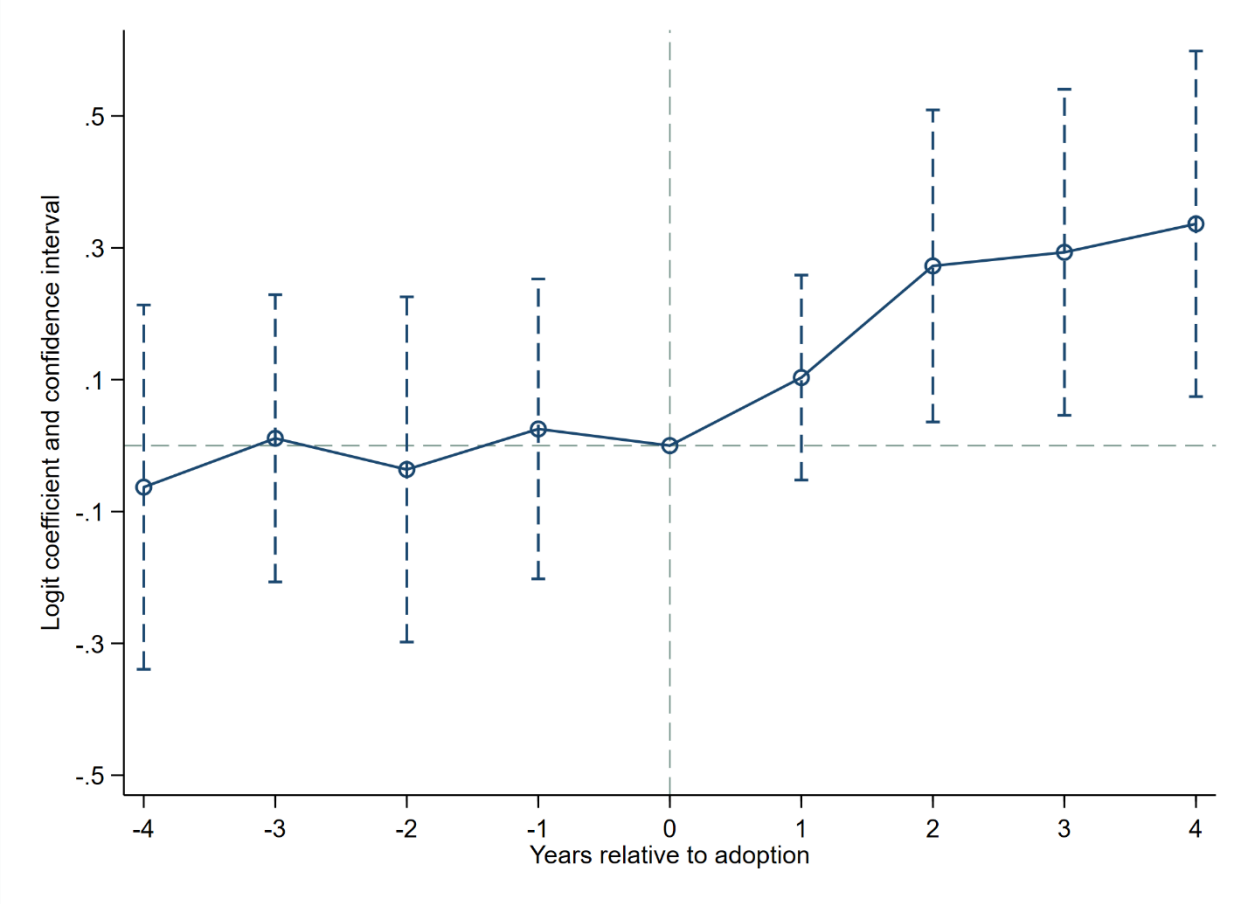


Figure 3. Dynamic Impact of Good Faith Adoption on Zero Leverage

This figure plots the point estimates together with 95% confidence interval of the impact of Good Faith adoption laws on zero leverage propensity in a four-year window around the adoption year. Specifically, the estimates are the outcome of running the following logit regression:

$$ZL_{ist} = \alpha + \beta_1 GF_{st}^{\leq -4} + \beta_2 GF_{st}^{-3} + \beta_3 GF_{st}^{-2} + \beta_4 GF_{st}^{-1} + \beta_5 GF_{st}^{+1} + \beta_6 GF_{st}^{+2} + \beta_7 GF_{st}^{+3} + \beta_8 GF_{st}^{\geq +4} + \text{Firm Controls}_{ist} + \text{State Controls}_{st} + \text{State Fixed effects} + \text{Year Fixed Effects} + \text{Industry Fixed Effects} + \varepsilon_{ist}$$

where the dependent variable is an indicator whether the firm has zero leverage or not (yes = 1; no = 0). GF_{st}^{L} for positive L (negative L) gets 1 if the firm is headquartered in a state that has adopted the law L years ago (will adopt the law in L years), and zero otherwise. We exclude the dummy variable corresponding to the year of law adoption, therefore all the β coefficients are estimated relative to the reference year, the year of adoption of good faith law. Standard errors are clustered at the state-level.

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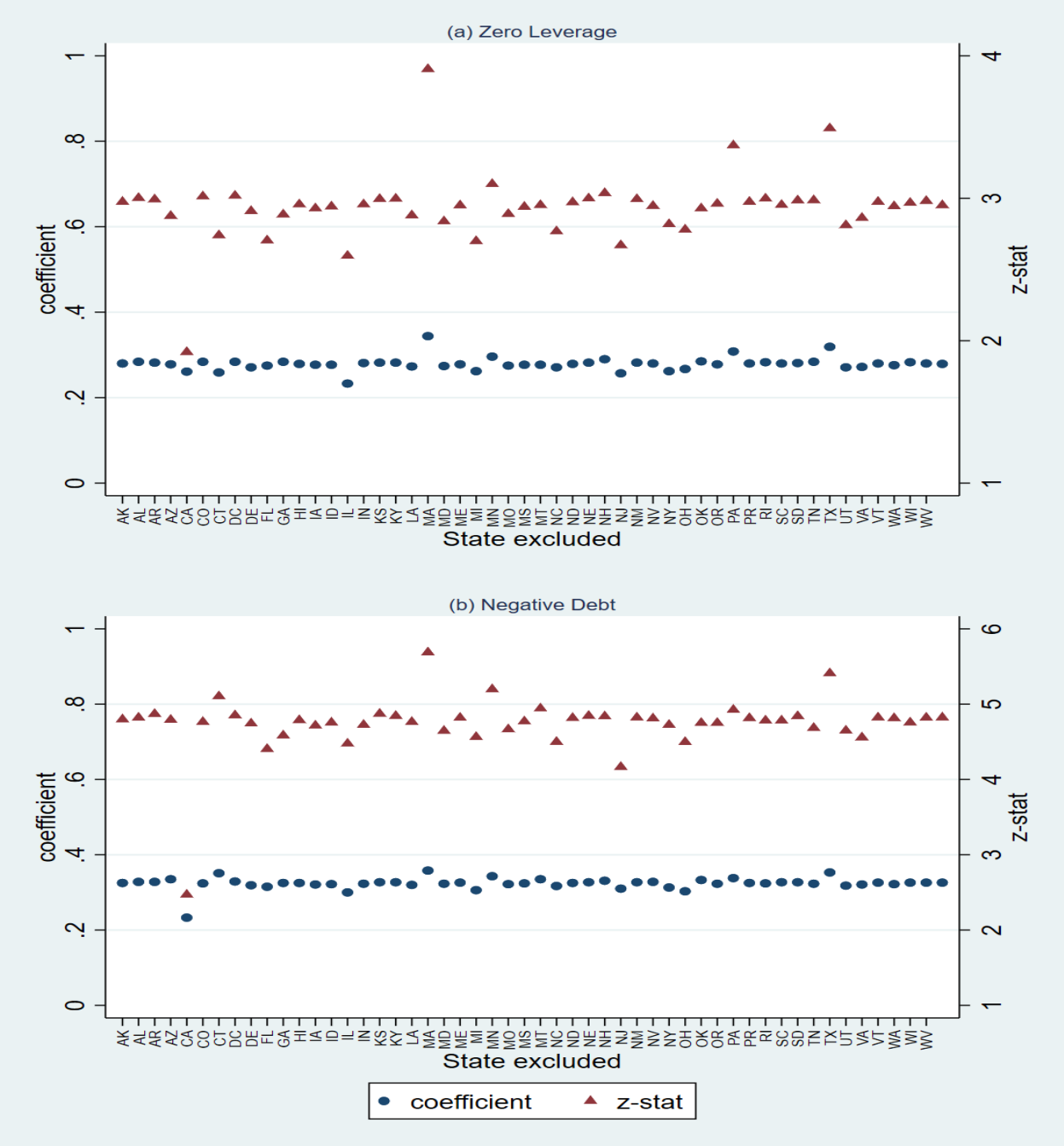


Figure 4: Employment Protection and Zero Leverage: Dropping States

This figure plots estimated coefficients (circles) and their z-stat (triangles) on the Good Faith indicator in regression specifications similar to model 5 from Table 4 but with 50 (instead of 51) states. We run 51 such regressions (by excluding one state at a time) and report the resulting estimates in this figure. Panel (a) reports the estimated numbers when Zero Leverage is used as dependent variable while Panel (b) reports the corresponding numbers for Negative Debt.

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Table 1. Descriptive Statistics

Variable	N Obs	Mean	Std. Dev.	Q 1	Median	Q 3
Dependent Variables:						
Zero Leverage	136,814	0.182	0.386	0.000	0.000	0.000
Negative Debt	136,814	0.277	0.448	0.000	0.000	1.000
Explanatory Variables:						
Good Faith	136,814	0.168	0.374	0.000	0.000	0.000
Control Variables:						
Implied Contract	136,814	0.509	0.500	0.000	1.000	1.000
Public Policy	136,814	0.516	0.500	0.000	1.000	1.000
Assets (in million USD)	136,586	522.257	3,685.046	8.218	34.821	158.811
Market-to-Book	106,688	2.367	3.789	0.791	1.427	2.670
Profitability	135,981	-0.031	0.273	-0.016	0.039	0.077
Capital Intensity	136,268	0.331	0.230	0.152	0.281	0.466
Dividend Payer	136,312	0.400	0.490	0.000	0.000	1.000
Cash Flow Volatility	108,189	0.081	0.122	0.018	0.035	0.081
Capex	133,467	0.079	0.084	0.026	0.054	0.101
Age (in years)	136,814	7.065	7.665	0.000	5.000	11.000
Working Capital	129,399	0.142	0.270	0.013	0.162	0.313
State GDP Growth	136,620	0.080	0.037	0.056	0.078	0.102
State per capita GDP	136,620	9.564	0.575	9.106	9.703	10.058
Circuit States' Good Faith	136,814	0.063	0.164	0.000	0.000	0.000
Non-Union	136,620	0.791	0.089	0.721	0.790	0.870
Industry Volatility	134,478	0.075	0.065	0.029	0.055	0.103
Industry Concentration	136,814	0.453	0.279	0.223	0.374	0.635
Employee Intensity	124,301	0.919	3.183	0.066	0.213	0.662

This table presents descriptive statistics for a sample of 136,814 firm observations between 1967 and 1995. Zero Leverage is an indicator set to one if the sum of book value of long-term debt (*dltt*) and short-term debt in current liabilities (*dlc*) is less than 5 percent of total assets, and zero otherwise. Negative Debt is an indicator variable set to one if the sum of book value of long-term debt (*dltt*) and short-term debt in current liabilities (*dlc*) less cash and cash equivalents (*che*) is negative, and zero otherwise. *Good Faith* is a dummy variable equaling one if the state has adopted a good faith exception in the year of analysis, zero otherwise. *Implied Contract* is a dummy variable equaling one if the state has adopted an implied contract exception in the year of analysis, zero otherwise. *Public Policy* is a dummy variable equaling one if the state has adopted a public policy exception in the year of analysis, zero otherwise. *Assets* is the book value of assets. *Market-to-Book* is the market value of equity (price per share at the end of the fiscal year multiplied by the number of shares outstanding) divided by the book value of equity. *Profitability* is operating cash flow scaled by net assets (total assets minus cash and cash equivalents). *Capital Intensity* is the ratio of property plant and equipment scaled by total assets. *Dividend Payer* is an indicator variable set to one if a firm pays a common dividend during a fiscal year and zero otherwise. *Cash Flow Volatility* is the standard deviation of the ratio of operating cash flow scaled by total assets over the preceding five years with a minimum of three years of data. *Capex* is capital expenditure scaled by total assets. *Age* is the number of years the firm is available in Compustat. *Working Capital* is operating current assets (current assets minus cash and cash equivalents) minus operating current liabilities (current liabilities minus short-term debt) scaled by total assets. *State GDP Growth* is the state-level GDP growth rate over the prior year. *State per capita GDP* is a state's GDP divided by its total population (in thousands). *Circuit States' Good Faith* is the fraction of other states in the same federal circuit region as the firm's headquarters state that have passed the good faith exception by year *t*. *Non-Union* is the average annual proportion of employees in each state that is not unionized. *Industry Volatility* is the annual average *Cash Flow Volatility* per industry and per state. *Industry Concentration* is the annual industry Herfindahl concentration index measured per state. *Employee Intensity* is the ratio of employees (in thousands) over total assets (in mln. USD).

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Table 2. Proportion of Zero Leverage Firms by State

Panel A: Proportion of Zero Leverage Firms per State

		Zero leverage		Negative debt	
State	Good Faith Adopting	Before GF	After GF Adoption	Before GF	After GF Adoption
AK	Yes	0.148	.	0.222	0.087
AL	No	0.151	0.151	0.280	0.280
AR	No	0.050	0.050	0.136	0.136
AZ	Yes	0.147	0.188	0.281	0.352
CA	Yes	0.161	0.290	0.288	0.497
CO	No	0.239	0.239	0.410	0.410
CT	Yes	0.102	0.163	0.224	0.322
DC	No	0.145	0.145	0.286	0.286
DE	Yes	0.091	0.263	0.201	0.425
FL	No	0.167	0.167	0.326	0.326
GA	No	0.158	0.158	0.293	0.293
HI	No	0.112	0.112	0.199	0.199
IA	No	0.129	0.129	0.280	0.280
ID	Yes	0.238	0.177	0.410	0.405
IL	No	0.173	0.173	0.274	0.274
IN	No	0.172	0.172	0.297	0.297
KS	No	0.119	0.119	0.270	0.270
KY	No	0.136	0.136	0.283	0.283
LA	No	0.138	0.138	0.255	0.255
MA	Yes	0.163	0.264	0.331	0.483
MD	No	0.190	0.190	0.343	0.343
ME	No	0.296	0.296	0.489	0.489
MI	No	0.169	0.169	0.290	0.290
MN	No	0.214	0.214	0.370	0.370
MO	No	0.140	0.140	0.244	0.244
MS	No	0.147	0.147	0.230	0.230
MT	Yes	0.033	0.250	0.367	0.306
NC	No	0.152	0.152	0.276	0.276
ND	No	0.038	0.038	0.094	0.094
NE	No	0.225	0.225	0.340	0.340
NH	Yes	0.223	0.228	0.223	0.394
NJ	No	0.197	0.197	0.368	0.368
NM	No	0.134	0.134	0.345	0.345
NV	Yes	0.191	0.216	0.315	0.360
NY	No	0.169	0.169	0.347	0.347
OH	No	0.129	0.129	0.228	0.228
OK	Yes	0.134	0.190	0.234	0.310
OR	No	0.168	0.168	0.332	0.332
PA	No	0.148	0.148	0.267	0.267
PR	No	0.120	0.120	0.219	0.219
RI	No	0.147	0.147	0.290	0.290
SC	No	0.150	0.150	0.229	0.229
SD	No	0.118	0.118	0.200	0.200
TN	No	0.122	0.122	0.229	0.229
TX	No	0.162	0.162	0.296	0.296
UT	Yes	0.214	0.298	0.391	0.537
VA	No	0.160	0.160	0.291	0.291
VT	No	0.126	0.126	0.370	0.370
WA	No	0.227	0.227	0.389	0.389
WI	No	0.170	0.170	0.284	0.284
WV	No	0.058	0.058	0.175	0.175
WY	Yes	0.113	0.250	0.270	0.250

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Panel B: Differences in Zero Leverage Proportions per State

Proportion of firms with:	**Zero leverage**	**Negative debt**
Good Faith adopting states:		
Before adoption of GF	14.7%	28.1%
After adoption of GF	20.8%	36.6%
Non Good Faith adopting states:	14.6%	27.1%
Significance levels (p-value):		
Diff before vs after	*0.000*	*0.000*
Diff before vs non-adopters	*0.814*	*0.378*
Diff after vs non-adopters	*0.000*	*0.000*

This table shows percentage proportions of firms with zero leverage and negative debt per state in Panel A. For states that adopt the good faith exception at some point in the sample period of 1967-1995, proportions are shown before and after the adoption. Panel B shows the average proportions of zero leverage and negative debt firms in good faith adopting states versus non-adopting states. For the adopting states, proportions are shown before and after the adoption. At the bottom of the table, results from two-tailed t-tests are shown, which test for (1) the difference between the proportion of zero leverage and negative debt firms in good faith adopting states before versus after the adoption (Diff before vs after); (2) the difference between non-adopting states and adopting states before the adoption of the good faith exception (Diff before vs non-adopters) and (3) the difference between non-adopting states and adopting states after the adoption of the good faith exception (Diff after vs non-adopters).

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Table 3. What Predicts the Timing of States' Good Faith Exception Law

	(1)	(2)	(3)
Zero Leverage	4.474	2.774	3.430
	(1.419)	(1.187)	(1.057)
State GDP Growth	5.921	2.646	2.981
	(0.451)	(0.221)	(0.232)
State per Capita GDP	-2.009	-3.831	-1.342
	(-0.845)	(-1.629)	(-0.610)
Right-to-Work Laws	0.373	0.435	0.386
	(0.499)	(0.402)	(0.408)
Implied Contract	0.128	-0.212	-0.204
	(0.174)	(-0.210)	(-0.232)
Public Policy	2.233	2.903*	2.762*
	(1.574)	(1.669)	(1.780)
State Union Membership		-0.020	-0.054
		(-0.487)	(-1.208)
Change in State Union Membership		-0.112	-0.173
		(-0.938)	(-1.308)
Circuit States' Good Faith		3.185*	3.256**
		(1.930)	(2.163)
Unemployment Rate		-0.158	-0.023
		(-0.810)	(-0.130)
Change in Unemployment Rate		0.133	-0.173
		(0.332)	(-0.323)
year effects	no	no	yes
Observations	1,253	1,253	1,253
Log likelihood	-14.21	-23.21	-11.80

This table shows regression results from a Cox proportional hazard model explaining the incidence (hazard) of a state adoption the good faith exception between 1967 and 1995. A *failure event* occurs when a certain state adopts the good faith exception. States are dropped from the sample when the *failure event* happens. *Zero Leverage* is the proportion of firms with zero leverage in a particular state and year. The table has three columns. In Columns (1) and (2) various state-level variables are introduced. In Column (3), year effects are added. See appendix A.3 for variable definitions. ***, **, * denote significance at 1%, 5%, and 10% respectively.

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Table 4. Employment Protection and Zero Leverage

Panel A: Dependent variable is *Zero Leverage*

	(1)	(2)	(3)	(4)	(5)
Good Faith	0.195***	0.274***	0.261***	0.263***	0.280***
	(3.243)	(3.682)	(3.211)	(2.959)	(2.975)
Implied Contract	0.016	0.039	0.041	0.055	0.051
	(0.301)	(0.505)	(0.460)	(0.615)	(0.635)
Public Policy	-0.079	-0.072	-0.062	-0.040	-0.026
	(-1.542)	(-1.246)	(-0.859)	(-0.512)	(-0.394)
Log Assets		-0.375***	-0.367***	-0.349***	-0.348***
		(-17.785)	(-17.217)	(-12.996)	(-13.048)
Market-to-Book		0.022***	0.024***	0.020***	0.020***
		(4.953)	(5.698)	(3.997)	(4.027)
Profitability		1.322***	1.373***	1.490***	1.495***
		(11.699)	(10.777)	(9.223)	(9.168)
Capital Intensity		-3.250***	-3.176***	-3.771***	-3.792***
		(-14.973)	(-14.117)	(-15.236)	(-15.317)
Dividend Payer		0.798***	0.862***	0.930***	0.929***
		(9.548)	(10.143)	(13.985)	(14.082)
Cash Flow Volatility			0.107	-0.137	-0.145
			(0.467)	(-0.770)	(-0.819)
Capex				1.709***	1.775***
				(6.625)	(6.983)
Log Age				-0.098**	-0.099**
				(-2.113)	(-2.134)
Working Capital				-0.526***	-0.529***
				(-3.555)	(-3.529)
State GDP Growth					-2.107***
					(-3.308)
State per capita GDP					-0.074
					(-0.196)
Circuit States' Good Faith					0.071
					(0.462)
Constant Term	-2.367***	-0.561	-0.984**	-0.745	-0.012
	(-5.369)	(-1.234)	(-1.986)	(-1.423)	(-0.004)
Industry Effects	yes	yes	yes	yes	yes
State Effects	yes	yes	yes	yes	yes
Year Effects	yes	yes	yes	yes	yes
Cluster	state	state	state	state	state
Observations	136,814	106,139	91,610	87,430	87,328
Log likelihood	-61495	-43314	-36246	-34691	-34647
Pseudo R-squared	0.0522	0.156	0.150	0.155	0.155

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Panel B: Dependent variable is *Negative Debt*

	(1)	(2)	(3)	(4)	(5)
Good Faith	0.267***	0.313***	0.302***	0.311***	0.325***
	(-4.298)	(4.922)	(4.916)	(4.448)	(4.798)
Implied Contract	-0.088*	-0.064	-0.078	-0.065	-0.064
	(-1.897)	(-1.173)	(-1.177)	(-0.958)	(-1.121)
Public Policy	-0.064	-0.032	-0.007	0.004	0.008
	(-1.164)	(-0.446)	(-0.089)	(0.050)	(0.112)
Log Assets		-0.324***	-0.295***	-0.288***	-0.287***
		(-11.311)	(-9.934)	(-7.808)	(-7.802)
Market-to-Book		0.038***	0.037***	0.033***	0.033***
		(9.085)	(9.539)	(7.771)	(7.809)
Profitability		1.295***	1.526***	1.980***	1.983***
		(11.676)	(9.483)	(8.529)	(8.499)
Capital Intensity		-3.508***	-3.408***	-4.405***	-4.418***
		(-23.992)	(-21.108)	(-22.612)	(-22.812)
Dividend Payer		0.764***	0.828***	0.917***	0.918***
		(9.123)	(9.575)	(13.222)	(13.176)
Cash Flow Volatility			0.659***	0.043	0.036
			(2.783)	(0.284)	(0.245)
Capex				2.046***	2.101***
				(8.771)	(9.039)
Log Age				-0.101**	-0.102**
				(-2.252)	(-2.275)
Working Capital				-1.556***	-1.558***
				(-7.302)	(-7.259)
State GDP Growth					-1.616***
					(-2.804)
State per capita GDP					-0.209
					(-0.502)
Circuit States' Good Faith					0.014
					(0.068)
Constant Term	-1.945***	0.824**	0.434	1.026**	2.848
	(-6.352)	(2.486)	(1.230)	(2.573)	(0.841)
Industry Effects	yes	yes	yes	yes	yes
State Effects	yes	yes	yes	yes	yes
Year Effects	yes	yes	yes	yes	yes
Cluster	state	state	state	state	state
Observations	136,814	106,139	91,610	87,430	87,328
Log likelihood	-75508	-52889	-44986	-42368	-42315
Pseudo R-squared	0.065	0.172	0.162	0.178	0.178

This table presents coefficients and corresponding z-statistics of logistic regressions examining the relation between employment protection laws and the likelihood of zero leverage and negative debt. The dependent variable in Panel A is Zero Leverage. The dependent variable in Panel B is Negative Debt. Each panel has five columns. Column (1) is the base model without additional control variables. Columns (2), (3) and (4) progressively introduce various firm-specific characteristics. Column (5) introduces state-specific variables. All regressions include industry, state and year effects. Residuals are clustered by state. See appendix A.3 for variable definitions. ***, **, * denote significance at 1%, 5%, and 10% respectively.

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Table 5. Employment Protection and Average Firm Leverage: The Role of Zero Leverage Firms

	(1)	(2)	(3)	(4)
Dep. Var.:	**Book Leverage**		**Market Leverage**	
Good Faith	-0.010*	-0.002	-0.010*	-0.002
	(0.006)	(0.005)	(0.005)	(0.004)
Zero Leverage or Negative Debt		-0.189***		-0.185***
		(0.005)		(0.004)
Implied Contract	-0.002	-0.004*	-0.003	-0.005
	(0.003)	(0.002)	(0.006)	(0.006)
Public Policy	0.004	0.003	0.006	0.005
	(0.004)	(0.003)	(0.005)	(0.006)
Log Assets	0.032***	0.022***	0.053***	0.043***
	(0.003)	(0.002)	(0.003)	(0.002)
Market-to-Book	-0.002***	-0.002***	-0.007***	-0.007***
	(0.000)	(0.000)	(0.001)	(0.001)
Profitability	-0.181***	-0.138***	-0.194***	-0.152***
	(0.017)	(0.013)	(0.024)	(0.019)
Capital Intensity	0.210***	0.064***	0.222***	0.079***
	(0.015)	(0.012)	(0.014)	(0.013)
Cash Flow Volatility	0.157***	0.147***	0.060**	0.049**
	(0.027)	(0.025)	(0.025)	(0.020)
Working Capital	-0.242***	-0.266***	-0.099***	-0.122***
	(0.015)	(0.011)	(0.012)	(0.009)
Log Age	0.005*	0.004**	0.024***	0.023***
	(0.003)	(0.002)	(0.005)	(0.005)
Observations	88,409	88,409	88,409	88,409
Adj. R-squared	0.657	0.726	0.682	0.732

This table presents coefficients and corresponding t-statistics of panel regressions examining the relation between employment protection laws and the average firm leverage. The dependent variable in Columns 1 and 2 is book leverage whereas in the last two columns we use market leverage as the dependent variable. In Columns 2 and 4, we also include an indicator variable that gets the value 1 if the firm in that specific year is zero leverage or have negative net debt, and zero otherwise. All regressions include firm and year fixed effects. Residuals are clustered by state.

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Table 6. Heterogeneous Effect of Employment Protection

Dependent Variable:	**Zero Leverage**		**Negative Debt**	
	(1)	(2)	(3)	(4)
Panel A: Unionization				
	Low Unionization	*High Unionization*	*Low Unionization*	*High Unionization*
Good Faith	0.273***	0.211	0.338***	0.225
	(2.861)	(1.114)	(4.195)	(1.538)
Wald test Prob>F	0.046		0.000	
Panel B: Industry Volatility				
	High Volatility	*Low Volatility*	*High Volatility*	*Low Volatility*
Good Faith	0.308**	0.037	0.418***	0.055
	(0.125)	(0.148)	(0.118)	(0.082)
Wald test Prob>F	0.017		0.001	
Panel C: Industry Concentration				
	Low Concentration	*High Concentration*	*Low Concentration*	*High Concentration*
Good Faith	0.292***	0.194	0.333***	0.251***
	(2.593)	(1.264)	(2.955)	(2.761)
Wald test Prob>F	0.003		0.039	
Panel D: Employee Intensity				
	High Intensity	*Low Intensity*	*High Intensity*	*Low Intensity*
Good Faith	0.365**	0.153*	0.489***	0.133
	(2.007)	(1.754)	(3.557)	(1.522)
Wald test Prob>F	0.038		0.000	

This table presents coefficients and corresponding z-statistics of logistic regressions examining the relation between employment protection laws and the likelihood of zero leverage and negative debt, utilizing sub-samples based on unionization rates measured at the state-year level, industry volatility, industry concentration and firms' labor intensity. The table has four columns: the first two columns utilize *Zero Leverage* as a dependent variable and the latter two use *Negative Debt.* All regressions include (not reported) the full set of control variables and fixed effects as per Table 4 Column 5, and residuals are clustered by state. See appendix A.3 for variable definitions. ***, **, * denote significance at 1%, 5%, and 10% respectively.

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Table 7. Alternative Regression Methods

	(1)	(2)	(3)	(4)
Regression Method:	Linear Probability Model		Probit	
Dependent Variable:	Zero Leverage	Negative Debt	Zero Leverage	Negative Debt
Good Faith	0.043***	0.061***	0.159***	0.187***
	(3.923)	(4.694)	(3.095)	(5.012)
Implied Contract	0.005	-0.011	0.031	-0.034
	(0.435)	(-1.043)	(0.696)	(-1.017)
Public Policy	-0.007	-0.002	-0.020	-0.001
	(-0.675)	(-0.195)	(-0.536)	(-0.021)
Log Assets	-0.041***	-0.044***	-0.195***	-0.165***
	(-20.424)	(-9.755)	(-14.898)	(-8.427)
Market-to-Book	0.003***	0.006***	0.012***	0.020***
	(4.922)	(9.579)	(4.509)	(8.347)
Profitability	0.224***	0.345***	0.822***	1.081***
	(10.850)	(11.832)	(10.269)	(10.285)
Capital Intensity	-0.433***	-0.668***	-2.002***	-2.477***
	(-16.598)	(-27.106)	(-16.726)	(-25.725)
Dividend Payer	0.111***	0.143***	0.517***	0.529***
	(15.721)	(14.932)	(14.651)	(13.749)
Cash Flow Volatility	0.066**	0.073**	0.011	0.063
	(2.283)	(2.549)	(0.103)	(0.699)
Capex	0.106***	0.213***	0.862***	1.108***
	(3.902)	(5.823)	(6.891)	(8.460)
Log Age	-0.013*	-0.019**	-0.052**	-0.061**
	(-1.796)	(-2.363)	(-1.980)	(-2.269)
Working Capital	-0.106***	-0.291***	-0.310***	-0.864***
	(-4.936)	(-8.940)	(-3.772)	(-7.926)
State GDP Growth	-0.281***	-0.256***	-1.175***	-0.914***
	(-3.332)	(-2.809)	(-3.430)	(-2.914)
State per capita GDP	-0.017	-0.039	-0.043	-0.104
	(-0.308)	(-0.507)	(-0.209)	(-0.440)
Circuit States' Good Faith	-0.001	-0.002	0.040	0.007
	(-0.056)	(-0.054)	(0.470)	(0.061)
Year Fixed Effects	yes	yes	yes	yes
Firm Fixed Effects	yes	yes	yes	yes
State Fixed Effects	yes	yes	yes	yes
Observations	87,328	87,328	87,328	87,328
R-squared	0.136	0.193	0.154	0.176
Log likelihood	-		-34697	-42383

This table presents the results of our full model regression specification (similar to those presented in Column 5 Table 4) using alternative regression methods. The first two columns show the results using a linear probability model while the second two columns provide estimates of a probit regression. Standard errors are clustered at the state level. See appendix A.3 for variable definitions. ***, **, * denote significance at 1%, 5%, and 10% respectively.

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Table 8. The Impact of (Time-varying) Industry Factors

Dependent Variable:	(1)	(2)	(3)	(4)	(5)	(6)
	Zero Leverage			Negative Debt		
Good Faith	0.044***	0.043***	0.035***	0.066***	0.061***	0.040***
	(4.152)	(3.923)	(3.441)	(4.818)	(4.694)	(3.975)
Implied Contract	0.005	0.005	0.004	-0.011	-0.011	-0.011
	(0.429)	(0.435)	(0.387)	(-0.939)	(-1.043)	(-1.373)
Public Policy	-0.008	-0.007	-0.004	-0.004	-0.002	-0.001
	(-0.823)	(-0.675)	(-0.512)	(-0.327)	(-0.195)	(-0.071)
Log Assets	-0.040***	-0.041***	-0.040***	-0.044***	-0.044***	-0.043***
	(-16.123)	(-20.424)	(-21.911)	(-8.316)	(-9.755)	(-10.213)
Market-to-Book	0.004***	0.003***	0.003***	0.008***	0.006***	0.006***
	(6.492)	(4.922)	(4.679)	(11.799)	(9.579)	(9.496)
Profitability	0.242***	0.224***	0.228***	0.361***	0.345***	0.349***
	(12.207)	(10.850)	(10.696)	(11.989)	(11.832)	(11.441)
Capital Intensity	-0.393***	-0.433***	-0.437***	-0.639***	-0.668***	-0.675***
	(-11.859)	(-16.598)	(-17.468)	(-18.383)	(-27.106)	(-27.903)
Dividend Payer	0.106***	0.111***	0.111***	0.135***	0.143***	0.141***
	(12.459)	(15.721)	(15.087)	(11.454)	(14.932)	(14.313)
Cash Flow Volatility	0.108***	0.066**	0.052*	0.140***	0.073**	0.053*
	(3.917)	(2.283)	(1.720)	(4.784)	(2.549)	(1.775)
Capex	0.171***	0.106***	0.120***	0.297***	0.213***	0.236***
	(5.465)	(3.902)	(4.183)	(6.785)	(5.823)	(6.239)
Log Age	-0.015*	-0.013*	-0.013*	-0.022**	-0.019**	-0.020**
	(-1.935)	(-1.796)	(-1.838)	(-2.409)	(-2.363)	(-2.460)
Working Capital	-0.159***	-0.106***	-0.102***	-0.341***	-0.291***	-0.287***
	(-7.204)	(-4.936)	(-4.548)	(-10.224)	(-8.940)	(-8.381)
State GDP Growth	-0.335***	-0.281***	-0.152	-0.319***	-0.256***	-0.149
	(-3.886)	(-3.332)	(-1.610)	(-3.198)	(-2.809)	(-1.514)
State per capita GDP	0.010	-0.017	-0.021	-0.006	-0.039	-0.018
	(0.170)	(-0.308)	(-0.435)	(-0.064)	(-0.507)	(-0.267)
Circuit States' Good Faith	0.006	-0.001	-0.001	0.013	-0.002	-0.004
	(0.252)	(-0.056)	(-0.026)	(0.308)	(-0.054)	(-0.120)
Industry-Year FE	no	no	yes	no	no	yes
Industry FE	no	yes	no	no	yes	no
State FE	yes	yes	yes	yes	yes	yes
Year FE	yes	yes	no	yes	yes	no
Observations	87,328	87,328	87,328	87,328	87,328	87,328
Adj-R-squared	0.117	0.136	0.149	0.166	0.193	0.207

This table presents coefficients and corresponding t-statistics of linear probability models examining the relation between employment protection laws and the likelihood of zero leverage and negative debt. The dependent variable in the first three columns is Zero Leverage, whereas the dependent variable for the last three columns is negative net debt. We have three specifications per dependent variable. The first one includes only state and year fixed effects, the second adds industry fixed effects, and the third specification has the full set of industry-year fixed effects and as a result does not need separate year and industry fixed effects. Residuals are clustered by state. See appendix A.3 for variable definitions. ***, **, * denote significance at 1%, 5%, and 10% respectively.

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Table 9. Within-firm Changes in Zero Leverage Policies

Dependent Variable:	(1) Zero Leverage	(2) Zero Leverage	(3) Negative Debt	(4) Negative Debt
Good Faith	0.014**	0.019**	0.030***	0.026**
	(2.209)	(2.165)	(4.247)	(2.271)
Implied Contract	0.009	0.010	-0.012**	-0.011*
	(1.270)	(1.441)	(-2.060)	(-1.910)
Public Policy	-0.004	-0.003	-0.003	-0.002
	(-0.475)	(-0.392)	(-0.411)	(-0.273)
Log Assets	-0.054***	-0.057***	-0.056***	-0.055***
	(-14.006)	(-14.910)	(-9.410)	(-9.704)
Market-to-Book	-0.000	-0.000	0.001*	0.001
	(-0.534)	(-1.023)	(1.691)	(1.491)
Profitability	0.121***	0.120***	0.225***	0.222***
	(7.416)	(7.629)	(8.707)	(8.713)
Capital Intensity	-0.471***	-0.479***	-0.792***	-0.813***
	(-15.180)	(-14.512)	(-26.934)	(-24.971)
Dividend Payer	0.043***	0.044***	0.068***	0.067***
	(9.731)	(8.442)	(13.976)	(12.022)
Cash Flow Volatility	-0.044	-0.049	-0.040	-0.050
	(-1.283)	(-1.341)	(-1.005)	(-1.231)
Capex	0.030*	0.030	0.032	0.039
	(1.714)	(1.517)	(1.390)	(1.645)
Log Age	-0.001	-0.000	-0.006	-0.007
	(-0.091)	(-0.035)	(-0.943)	(-1.123)
Working Capital	0.027	0.029	-0.150***	-0.150***
	(1.591)	(1.645)	(-5.325)	(-5.133)
State GDP Growth	-0.102	-0.065	-0.015	-0.015
	(-1.599)	(-0.978)	(-0.257)	(-0.254)
State per capita GDP	-0.036	-0.053	-0.080	-0.049
	(-0.763)	(-1.082)	(-1.646)	(-0.982)
Circuit States' Good Faith	0.031**	0.034**	0.006	0.012
	(2.367)	(2.353)	(0.270)	(0.560)
Industry*Year Fixed Effects	no	yes	no	yes
Firm Fixed Effects	yes	yes	yes	yes
State Fixed Effects	yes	yes	yes	yes
Observations	87,322	87,322	87,322	87,322
R-squared	0.043	0.061	0.077	0.098

This table presents coefficients and corresponding t-statistics of linear probability models examining the relation between employment protection laws and the likelihood of zero leverage and negative debt within firms. The dependent variable in the first two (second two) columns are Zero Leverage (Negative Debt). The difference between the two specifications per each dependent variable is the set of fixed effects that are included in regressions. The first specification includes firm, state, and year fixed effects while the second specification allows for time-varying industry fixed effects, which renders the year fixed effects redundant. Residuals are clustered by state. See appendix A.3 for variable definitions. ***, **, * denote significance at 1%, 5%, and 10% respectively.

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Table 10. Employment Protection and Zero Leverage: State-Level Specification

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable:	Zero Leverage			Negative Debt		
Good Faith	0.042***	0.045***	0.045***	0.063***	0.066***	0.066***
	(3.686)	(3.584)	(3.626)	(3.733)	(3.417)	(3.443)
Implied Contract	0.004	0.003	0.003	-0.008	-0.009	-0.009
	(0.527)	(0.428)	(0.440)	(-0.879)	(-0.953)	(-0.944)
Public Policy	-0.020**	-0.018*	-0.018*	-0.024**	-0.021	-0.021
	(-2.239)	(-1.792)	(-1.791)	(-2.053)	(-1.503)	(-1.501)
State GDP Growth		-0.229**	-0.234**		-0.224*	-0.230*
		(-2.466)	(-2.500)		(-1.781)	(-1.810)
Log per capital GDP		0.076	0.076		0.106	0.106
		(1.215)	(1.215)		(1.106)	(1.106)
Circuit States' Good Faith		0.022	0.023		0.031	0.031
		(0.834)	(0.837)		(0.801)	(0.804)
State Trends	no	no	yes	no	no	yes
State Fixed Effects	yes	yes	yes	yes	yes	yes
Year Fixed Effects	yes	yes	yes	yes	yes	yes
Cluster	state	state	state	state	state	state
Observations	1,476	1,476	1,476	1,476	1,476	1,476
Adj. R-squared	0.516	0.527	0.528	0.594	0.602	0.602

This table presents coefficients and corresponding t-statistics of OLS regressions examining the relation between employment protection laws and the proportion of zero leverage (Columns 1-3) and negative debt firms (Columns 4-6) in the states. The unit of observation is state-year. The regressions are weighted by the average number of firms in each state during the sample period. For each dependent variable, we have three specifications. The fist specification is the base model with state and year fixed effects but without additional control variables and state trends. The second specification introduces some other important state level controls, while the third specification adds specific state trend effects. Standard errors are clustered by state. ***, **, * denote significance at 1%, 5%, and 10% respectively.

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APPENDICES

Appendix A.1: Content of the Wrongful Discharge Laws

Good Faith Exception

Originates from contract (legal) law principle that no single party in an agreement can deny the other the full benefits of a contractual relationship. The exception requires employers to treat workers in a fair manner (i.e., in good faith) and not take actions that would deprive employees of the benefit of employment without "just cause". For example, employers cannot discharge workers out of bad faith, malevolence, or retaliation nor can they prematurely terminate a working relationship before the employee received a compensation (stock options; bonus) to which he is entitled to. While this exception is rather broad in nature, it has been applied narrowly once adopted in the respective States on issues like pension benefits, sales commission, sick leave or earned vacation time (Autor, Donohue and Schwab 2006).

Implied Contract Exception

Originates from the principle that an employer cannot discharge an employee with whom the employer has created and explicit/implicit promise not to terminate without good cause. These promises may be oral; if written in a handbook, they do not need to be negotiated with employees individually. Courts have also determined that employee tenure, a history of promotions or salary raises, general company policies, and typical industry practices can constitute an implied promise of ongoing employment.

Public Policy Exception

Originates from contract law principle that an employer who discharges an employee in violation of an important public policy, does so for a reason that the legislature/society deems so unreasonable that statutory or common law prohibits it (either expressly or impliedly) (Baucus and Dworking, 1998; Autor 2003). The exception is there to protect employees from being discharged for performing a public service even if the action is not in the employer's interest (e.g., performing jury duties, refusing illegal act participations, exercising workers' compensation claims, or reporting employer's misbehavior).

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Appendix A.2: State-Level Wrongful Discharge Laws

State	Good Faith Exception	Implied Contract Exception	Public Policy Exception
Alabama		1987	
Alaska	1983	1983	1986
Arizona	1985	1983 (reversed in 1984)	1985
Arkansas		1984	1980
California	1980	1972	1959
Colorado		1983	1985
Connecticut	1980	1985	1980
Delaware	1992		1992
Florida			
Georgia			
Hawaii		1986	1982
Idaho	1989	1977	1977
Illinois		1974	1978
Indiana		1987	1973
Iowa		1987	1985
Kansas		1984	1981
Kentucky		1983	1983
Louisiana	1998		
Maine		1977	
Maryland		1985	1981
Massachusetts	1977	1988	1980
Michigan		1980	1976
Minnesota		1983	1986
Mississippi		1992	1987
Missouri		1983 (reversed in 1988)	1985
Montana	1982	1987	1980
Nebraska		1983	1987
Nevada	1987	1983	1984
New Hampshire	1974 (reversed in 1980)	1988	1974
New Jersey		1985	1980
New Mexico			1983
New York		1982	
North Carolina			1985
North Dakota		1984	1987
Ohio		1982	1990
Oklahoma	1985 (reversed in 1989)	1976	1989
Oregon		1978	1975
Pennsylvania			1974
Rhode Island			
South Carolina		1987	1985
South Dakota		1983	1988
Tennessee		1981	1984
Texas		1985	1984
Utah	1989	1986	1989
Vermont		1985	1986
Virginia		1983	1985
Washington		1977	1984
West Virginia		1986	1978
Wisconsin		1985	1980
Wyoming	1994	1985	1989

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Appendix A.3: Variable Description and Measurement

Variable	Definition
Age	The number of years the firm is available in Compustat, calculated from 1995.
Assets	The value of total book assets in millions.
Capex	Capital expenditure scaled by total assets.
Capital Intensity	The amount of property, plant and equipment (tangible fixed assets) scaled by total assets.
Cash Flow Volatility	Standard deviation of the ratio of operating cash flow scaled by total assets over the preceding five years. Firms must at least have three years of data).
Circuit States' Good Faith	The fraction of other states in the same federal circuit region as the firm's headquarters state that have passed the good faith exception by year *t*.
Dividend Payer	An indicator variable set to one if a firm pays a common dividend during a fiscal year and zero otherwise.
Employee Intensity	The number of employees (in thousands) scaled by total assets.
Good Faith	An indicator variable set to one if the state in which a firm is headquartered has adopted the good faith exception by year *t* and zero otherwise.
Implied Contract	An indicator variable set to one if the state in which a firm is headquartered has adopted the implied contract exception by year *t* and zero otherwise.
Industry Concentration	Herfindahl concentration index, measured as the sum of squared relative sales, measured as each firm's sales divided by the industry's sales. The index is measured per industry (SIC 3 digit), state and year.
Industry Volatility	The average *Cash Flow Volatility* across all firms in the same industry (SIC 3 digit), state and year.
Market-to-Book	The market value of equity (price per share at the end of the fiscal year multiplied by the number of shares outstanding) divided by book value of equity.

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Negative Debt	An indicator variable set to one if the sum of book value of long-term debt (*dltt*) and short-term debt in current liabilities (*dlc*) less cash and cash equivalents (*che*) is negative, and zero otherwise.
Non-Union	One minus the annual state unionization rate (*Unionization*).
Profitability	Operating cash flow scaled by net assets (total assets minus cash and cash equivalents).
Public Policy	An indicator variable set to one if the state in which a firm is headquartered has adopted the public policy exception by year *t* and zero otherwise.
Right-to-Work Laws	An indicator variable set one if a firm is headquartered in a state that has passed right-to-work laws by year *t* and zero otherwise.
Sales	Total revenues of the company measured in millions.
State GDP Growth	The state-level GDP growth rate over the prior year.
State per capita GDP	A state's GDP (in thousands) divided by its total population.
Unemployment Rate	The annual unemployment rate per state.
Unionization	The average annual fraction of employees in each state that is unionized.
Working Capital	Operating current assets (current assets minus cash and cash equivalents) minus operating current liabilities (current liabilities minus short-term debt) scaled by total assets.
Zero Leverage	An indicator variable set to one if the sum of book value of long-term debt (*dltt*) and short-term debt in current liabilities (*dlc*) is less than 5 percent of total assets, and zero otherwise.

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Appendix A.4: Heterogeneous Effect of Labor Protection Laws: Interacting Variables

Panel A: Zero Leverage

Interaction variable:	**Non-Union**	**Industry Volatility**	**Industry Concentration**	**Employee Intensity**
	(1)	(2)	(3)	(4)
Good Faith	0.062	0.037	0.252**	0.125
	(0.120)	(0.148)	(0.104)	(0.126)
Good Faith x	0.244**	0.217**	0.036	0.167*
Interaction variable	(0.118)	(0.114)	(0.099)	(0.101)

Panel B: Negative Net Debt

Interaction variable:	**Non-Union**	**Industry Volatility**	**Industry Concentration**	**Employee Intensity**
	(1)	(2)	(3)	(4)
Good Faith	0.072	0.055	0.244***	0.171*
	(0.072)	(0.082)	(0.064)	(0.090)
Good Faith x	0.333***	0.319***	0.070	0.172*
Interaction variable	(0.071)	(0.093)	(0.086)	(0.091)

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